Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Computer Vision, Real-Time Inference, Model Serving | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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14th October 2026

Re: Machine Learning Engineer III, Data – ACV Auctions

Dear Hiring Manager,

I am applying for the Machine Learning Engineer III, Data role on the ML inspection team at ACV Auctions. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production under hard latency limits. At MAS Holdings I led computer vision systems for manufacturing, including care-label inspection that cut inspection time by 99.5% and raised operational efficiency by up to 300%. That is the same kind of problem as turning vehicle inspections into fast, reliable decisions.

Three parts of your posting match my work directly. For computer vision and damage detection, I built OpenCV-based defect and inspection systems at MAS, taking a care-label checker from a flatbed scanner to an industrial camera with a conveyor and PLC reject, and a loom-side fabric defect detector on a camera array with a few-second alert budget. For real-time inference, my published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline (Journal of the Audio Engineering Society, 2026), and I owned its full lifecycle from data acquisition to deployment and monitoring. For evaluation, I design benchmark suites with baselines and ablations, and I released four open datasets on Zenodo.

I should be direct about two gaps. My streaming and serving work has run on real-time ingestion and ETL pipelines, Airflow and Docker on AWS rather than Kafka or gRPC, and I have not used Kubeflow, Databricks or KServe in production. I have built the same pieces with other tools, and I expect to pick up your stack quickly. What draws me to ACV is that the models serve a high-volume marketplace, where precision, recall and throughput all matter and the work is judged by outcomes. Within the first months I would aim to contribute to damage detection quality and serving latency.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience in computer vision and real-time ML systems can support the ML inspection team at ACV Auctions. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD